

Phase 1 Pancreatic Cancer Trial Efficient LLM Document Generations

Kevin Kawchak | ORCID 0009-0007-5457-8667 | CEO ChemicalQDevice
10.5281/zenodo.xxxxxxxx | kevink@chemicalqdevice.com

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Independent research paper and practical adoption guide. It is not medical or regulatory advice and is not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor. The DOI placeholder 10.5281/zenodo.xxxxxxxx is filled at deposit. All figures derive from the author's repository sources and are illustrative unless tied to a cited reference.

Abstract

Phase 1 oncology trials advance only when a sequence of large regulatory and clinical documents is authored, reviewed, and approved. This paper shows how a repository based large language model (LLM), driven by a single master prompt that first writes and then executes a schedule of sub-prompts, can hasten the entire Phase 1 process by generating every relevant large document through an auditable mermaid to draft to full to final pipeline.

[DRAFTING INSTRUCTION] Write a 200-260 word abstract covering: (1) the patient-centered motivation (enrolled PDAC patients cannot wait for administrative slowdowns); (2) the method (repository LLM, single prompt, sub-prompt schedule, real-time auto-commit/auto-PR monitorability); (3) the six ACCELERATION document targets; (4) the verification stance (mermaid grounding, name-matching, GitHub monitorability); (5) the benefit-greater-than-risk conclusion and admin/prep time compression. State paper v1.0 and repository v4.2.0. Source the motivation and method from `inputs/llm-adoption/main.tex`, the document targets from `research/document-types/*`, and the workflow from `research/industry-workflow/*`.

[DRAFTING INSTRUCTION] Add the Keywords line (\keywords): Pancreatic ductal adenocarcinoma; Phase 1 clinical trial; large language model; document generation; repository based LLM; sub-prompt schedule; mermaid to LaTeX; clinical trial acceleration; daraxonrasib; verification before generation.

Keywords Pancreatic ductal adenocarcinoma; Phase 1 clinical trial; large language model; document generation; repository based LLM; sub-prompt schedule; mermaid to LaTeX; clinical trial acceleration; daraxonrasib; verification before generation.

1 Introduction

1.1 Patients Currently Enrolled in Trial Cannot Wait for Slowdowns

[DRAFTING INSTRUCTION] Open with OUTLINE theme 1. Make the patient-centered case: enrolled Phase 1 PDAC patients have under-13-percent five-year survival and cannot wait for administrative slowdowns; faster paperwork enables faster treatment; new document iterations now take 1-4 days; a single prompt grows a draft to a full to a final document. Render `mermaid/fig-17-patient-time-saved-cascade.md` and `fig-10-iteration-timeline.md` as TikZ figures. Source survival framing from `inputs/references.bib` (Siegel-class statistics via the protocol works) and the 1-4 day cadence from `prompts/prompt-paper.md` (OUTLINE).

1.2 Effective Verifications Prove Utility

[DRAFTING INSTRUCTION] *Introduce OUTLINE theme 2: grounding via Mermaid figures; figures match paper context; AI-generated files viewable on GitHub in real time for monitorability; paper repository and directory names match the repository names; paper file names match repo file names. Render mermaid/fig-15-name-matching-verification.md and fig-14-grounding-mermaid-to-tikz.md. Source from prompts/prompt-paper.md (OUTLINE 2) and inputs/llm-adoption/main.tex (monitorability).*

1.3 Patient Lives Extended

[DRAFTING INSTRUCTION] *Introduce OUTLINE theme 3: probable benefit greater than probable risk (state exactly how), and the mechanism by which the method compresses administrative/prep time. Render mermaid/fig-16-benefit-risk-framework.md and fig-11-time-bucket-compression.md. Source the three timeline buckets from research/document-types/Gemini-3-1-Pro-DocTypes-2026-06-26.md.*

1.4 Scope, Contributions, and Relationship to the Adoption Guide

[DRAFTING INSTRUCTION] *State that this paper builds on the Oncology Trial PI LLM Adoption Guide [1] (the paper template, paper v1.0) and turns its hands-on guidance into a worked Phase 1 PDAC document-generation study. List contributions: (1) a single-prompt mermaid-draft-full-final pipeline; (2) coverage of the six ACCELERATION targets; (3) 20+ grounded figures reproduced identically in LaTeX; (4) a real-time monitorable build. Note the real-world daraxonrasib PDAC trial thread [2, 3, 4] and the Physical AI framework [5]. State repository v4.2.0. Render mermaid/fig-01-single-prompt-pipeline.md and fig-21-daraxonrasib-document-thread.md.*

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2 Methods

2.1 Repository Based LLM and Input Curation

[DRAFTING INSTRUCTION] Describe the repository-based LLM method from *inputs/llm-adoption/main.tex* (Repository Setup, LLM Limitations): single input directory with subdirectories; pre-chunked sources with READMEs; total input under 5 MB; READMEs under 25K tokens; sectioned outputs as right-sized chunks. Render *mermaid/fig-12-repo-llm-architecture.md* as a TikZ figure. List the exact inputs: *inputs/llm-adoption*, *inputs/references.bib*, *research/document-types*, *research/industry-workflow*.

2.2 Single Prompt, Sub-Prompt Schedule, and Stages

[DRAFTING INSTRUCTION] Explain Process A (generate sub-prompts) and Process B (execute them sequentially) under one master prompt [1]. Render *mermaid/fig-02-subprompt-schedule.md* and *fig-01-single-prompt-pipeline.md*. Name the four sub-prompts in *sub-prompts/* and the four stages (*mermaid*, *draft-paper*, *full-paper*, *final-paper*). Note adaptation from the trial-protocol workflow (*trial-protocol/sub-prompts*).

2.3 Phase 1 Document Landscape and the Six Acceleration Targets

[DRAFTING INSTRUCTION] Summarize the large Phase 1 documents and the six ACCELERATION targets. Render *mermaid/fig-03-phase1-document-landscape.md*, *fig-04-acceleration-targets.md*, *fig-23-ind-irb-initial-package.md*, and *fig-24-clinical-hold-response.md*. Build a full-width table of the six targets with the gate type and the schedule-value note. Source from *research/document-types/ChatGPT-5-5-Thinking-Extended-DocTypes-2026-06-26.md* and *Gemini-3-1-Pro-DocTypes-2026-06-26.md*.

2.4 Document Decision-Gate Taxonomy

[DRAFTING INSTRUCTION] Explain hard gate / protocol-defined gate / decision gate. Render *mermaid/fig-05-document-gate-taxonomy.md*. Build a full-width table mapping each gate type to example documents and to whether faster authoring helps. Source from *research/document-types/ChatGPT-5-5-Thinking-Extended-DocTypes-2026-06-26.md*.

2.5 Before, During, and After Phase 1: Data and Document Workflow

[DRAFTING INSTRUCTION] Describe the before/during/after workflow. Render *mermaid/fig-06-data-collection-pipeline.md*, *fig-07-pretrial-authoring.md*, *fig-08-intratrial-authoring.md*, and *fig-09-posttrial-authoring.md*. Include EDC, eCRF, eCOA/ePRO, CTCAE, RECIST, SDV, database lock, eDMS (Veeva Vault), TransCelerate templates, ICH E3 CSR, ICH E2F DSUR. Source from *research/industry-workflow/**.

2.6 Figures, Tables, and Formatting Method

[DRAFTING INSTRUCTION] Explain the five-color palette and the Mermaid-to-TikZ fidelity method. Render *mermaid/fig-19-five-color-palette.md* and *fig-14-grounding-mermaid-to-tikz.md*. State the formatting strategies learned from *trial-protocol/final-protocol/publication* (full-width ragged-right tables, clearpage per section, vspace/hspace, no overlaps). Note the paper section architecture *mermaid/fig-18-paper-section-architecture.md*.

2.7 Real-Time Auto-Commit and Auto-PR Monitorability

[DRAFTING INSTRUCTION] Describe the auto-commit/auto-PR process. Render *mermaid/fig-13-autocommit-monitorability.md*. State Rule 6/7/8 commit discipline (one commit per file; second-to-last fixes errors; last does repository updates). Source from *prompts/prompt-paper.md* and *inputs/llm-adoption/main.tex*.

3 Results

3.1 A Single Prompt Produced Four Stage Outputs

[DRAFTING INSTRUCTION] Report the build outputs: 24 Mermaid figures, a draft scaffold, a full paper, and a final paper, each with *main.tex*, *paperstyle.sty*, *references.bib*, *sections/*, and an Overleaf zip. Build a full-width table of stage, directory, artifacts, and commit count. Render *mermaid/fig-01-single-prompt-pipeline.md*. Source from *sub-prompts/** and the four stage directories.

3.2 The Six Acceleration Targets Addressed

[DRAFTING INSTRUCTION] Report how each of the six ACCELERATION targets is addressed by the method, with the document composition and the binding clock for each (30-day IND review; 30-day clinical-hold response review; DLT window for cohort review; EOP2 scheduling; database lock for CSR/NDA-BLA). Render *fig-04-acceleration-targets.md*, *fig-23-ind-irb-initial-package.md*, *fig-24-clinical-hold-response.md*. Full-width table. Source from *research/document-types/**.

3.3 New Document Iterations in 1 to 4 Days

[DRAFTING INSTRUCTION] Report the 1-4 day iteration cadence versus the weeks-to-months baseline, and which timeline bucket is compressed. Render *fig-10-iteration-timeline.md* and *fig-11-time-bucket-compression.md*. Source from *research/document-types/Gemini-3-1-Pro-DocTypes-2026-06-26.md*.

3.4 Effective Verifications: Grounding and Name-Matching

[DRAFTING INSTRUCTION] Report the verification results (OUTLINE 2): mermaid grounding; figures match context; GitHub real-time monitorability; repository/directory name matching; file-name matching. Render *fig-14-grounding-mermaid-to-tikz.md*, *fig-15-name-matching-verification.md*, *fig-20-quality-verification-gates.md*. Build a full-width table of the five checks with a concrete example for each.

3.5 Prior Single-Prompt Multi-File Developments (Evidence)

[DRAFTING INSTRUCTION] Report the prior single-prompt multi-file outputs as evidence: the Phase 2 protocol directory [3] (*trial-phase-2*); the law-physical-ai-trials review [6], adoption [1], and enactment [7] directories; the single-prompt-bill auto-bill-02 [8] and auto-bill-01 [9] directories. Build a full-width table of repository, directory, output type, and DOI. Render *fig-21-daraxonrasib-document-thread.md*.

3.6 2025 Application Papers Establishing LLM Trust

[DRAFTING INSTRUCTION] Report the 2025 trial application papers as the evidence trail leading to this project: [10, 11, 12, 13, 14, 15, 16, 17, 18, 19]. Render *fig-22-author-trust-timeline.md*. Full-width chronology table (date, work, contribution to trust).

4 Discussion

4.1 Probable Benefit Exceeds Probable Risk

[DRAFTING INSTRUCTION] Argue exactly how benefit exceeds risk (OUTLINE 3): benefits are faster paperwork, faster treatment, compressed admin/prep time, and more figures for oversight; risks are LLM error, formatting artifacts, and hallucinated context, each mitigated by human-in-the-loop review, mermaid grounding, name-matching, and real-time monitorability. Render *mermaid/fig-16-benefit-risk-framework.md*. For enrolled patients who cannot wait, withholding the method has its own cost.

4.2 Mechanism: Compressing Administrative and Prep Time

[DRAFTING INSTRUCTION] Explain that only the administrative/prep bucket compresses; clinical/operational time and fixed regulatory review clocks do not. Map the compression to the six ACCELERATION targets and the white space between phases. Render *fig-11-time-bucket-compression.md* and *fig-17-patient-time-saved-cascade.md*. Source from *research/document-types/**.

4.3 Prior Developments as Evidence for This Project

[DRAFTING INSTRUCTION] Detail the prior developments as a progression: the 2024-2025 LLM trust works [10, 13, 18, 19]; the PDAC simulation and digital-twin works [4, 14, 15, 16, 17]; the national platform framework [5]; the Phase 1 [2] and Phase 2 [3] protocols; the law/bill single-prompt repositories [6, 1, 7, 8, 9]. Argue this paper is the natural next single-prompt application to the document-generation problem itself. Render *fig-22-author-trust-timeline.md*.

4.4 Why Trust the Method: Verification, Not Faith

[DRAFTING INSTRUCTION] Tie to verification-before-generation [8] and the document quality gates that prevent rework cycles (amendments, regulator questions, IRB revisions, extra review cycle). Render *fig-20-quality-verification-gates.md*. Politely raise the adoption-guide framing about increasingly complex and voluminous AI outputs and the limits of human-only review.

4.5 Relationship to the Real-World Daraxonrasib Trial

[DRAFTING INSTRUCTION] Connect to the daraxonrasib PDAC trial thread: the documents accelerated here are exactly those the Phase 1 [2] and Phase 2 [3] protocols require, with daraxonrasib (RMC-6236) as the drug candidate [4]. Render *fig-21-daraxonrasib-document-thread.md*.

5 Limitations and Future Work

[DRAFTING INSTRUCTION] Enumerate limitations adapted from *inputs/llm-adoption/main.tex* (LLM Limitations, Project Proficiency): no reliance on internet search for large projects; variable pdf-compiler access so some author formatting is required; total input under 5 MB; per-file token limits; deterministic-output caveats for high-dimensional data; server errors as quality signals; uneven training across fields; security-related access restrictions in biology and chemistry. State that the method does not change tumor biology, follow-up windows,

or fixed regulatory review clocks, and that documents must still pass human medical, statistical, and regulatory review.

[DRAFTING INSTRUCTION] State future work: extend to Phase 2-to-3 briefing and Phase 3 protocol [3]; integrate live EDC/eDMS data feeds; add automated VVUQ-style verification of generated documents [8]; broaden to glioblastoma and lung adenocarcinoma [12, 19]; and connect to the national platform and site-documentation pipelines [5]. Keep this section self-standing on its page; avoid single-word lines.

6 Conclusions

[DRAFTING INSTRUCTION] Conclude in 2-3 tight paragraphs: (1) a single repository-based LLM prompt, with a sub-prompt schedule and a mermaid-draft-full-final pipeline, can generate every relevant large Phase 1 document and hasten the entire Phase 1 process by compressing administrative/prep time; (2) the method is monitorable and verifiable in real time (mermaid grounding, name-matching, GitHub auto-commit), so probable benefit exceeds probable risk for enrolled PDAC patients who cannot wait; (3) the prior single-prompt multi-file developments [3, 6, 1, 7, 8, 9] and the 2025 application papers establish the LLM trust that makes this application credible. State paper v1.0 and repository v4.2.0, and the daraxonrasib PDAC trial link [2, 4]. Render `mermaid/fig-17-patient-time-saved-cascade.md` as the closing figure.

7 References and Back Matter

[DRAFTING INSTRUCTION] The reference list renders below through the `ieeetr` style from `references.bib`. Confirm every reference has a clickable URL and a clickable DOI URL, and that no link runs off the right side (`UrlBreaks`). The back matter must include, in order: Acknowledgments; Author and ORCID; Ethical Disclosures; Data Availability; Rights and Permissions (CC); and Cite This Article.

Acknowledgments

[DRAFTING INSTRUCTION] Acknowledge Claude Code Opus 4.8 (1M context) as the build model and Chat-GPT 5.5 (Thinking Extended) and Gemini 3.1 Pro for the four research sources and manual editing assistance. State paper v1.0 assembled by Kevin Kawchak.

Author and ORCID

[DRAFTING INSTRUCTION] Kevin Kawchak, CEO ChemicalQDevice. ORCID iD as a plain hyperlink (no logo, Rule 12): <https://orcid.org/0009-0007-5457-8667>. Correspondence: kevin@chemicalqdevice.com.

Ethical Disclosures

[DRAFTING INSTRUCTION] Independent research paper and adoption guide. No real patients were enrolled, no protected health information was used, and no IRB approval was sought or obtained. Not medical or regulatory advice; not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor.

Data Availability

[DRAFTING INSTRUCTION] All paper sources (mermaid, draft-paper, full-paper, final-paper) are in the public repository and archived on Zenodo under CC BY 4.0 at DOI 10.5281/zenodo.##### (placeholder filled at deposit).

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Cite This Article

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References

- [1] K. Kawchak, "Oncology trial pi llm adoption guide," June 2026. DOI: 10.5281/zenodo.20843290, URL: <https://doi.org/10.5281/zenodo.20843290>. Paper template; also law-physical-ai-trials/tree/main/adoption.
- [2] K. Kawchak, "On-premises llm-directed robotic pancreaticoduodenectomy with perioperative daraxonrasib (rmc-6236) in kras-mutated pancreatic ductal adenocarcinoma," June 2026. DOI: 10.5281/zenodo.20780121, URL: <https://doi.org/10.5281/zenodo.20780121>. physical-ai-oncology-trials/tree/main/trial-protocol.
- [3] K. Kawchak, "A phase 2, daraxonrasib + llm guided robotic pdac whipple procedure, clinical trial protocol," June 2026. DOI: 10.5281/zenodo.20807027, URL: <https://doi.org/10.5281/zenodo.20807027>. physical-ai-oncology-trials/tree/main/trial-phase-2.
- [4] K. Kawchak, "2030: 60 second pancreatic ductal adenocarcinoma robotic whipple procedure and daraxonrasib simulation," May 2026. DOI: 10.5281/zenodo.20196639, URL: <https://doi.org/10.5281/zenodo.20196639>.
- [5] K. Kawchak, "National platform for physical ai oncology trials," Mar. 2026. DOI: 10.5281/zenodo.19244918, URL: <https://doi.org/10.5281/zenodo.19244918>.
- [6] K. Kawchak, "From h. r. 9510 to federal law: A narrative case for verified physical ai oncology trials," June 2026. DOI: 10.5281/zenodo.20685379, URL: <https://doi.org/10.5281/zenodo.20685379>. law-physical-ai-trials/tree/main/review.
- [7] K. Kawchak, "Earning the congress's vote: A new oncology trial framework for enacting h. r. 9510," June 2026. DOI: 10.5281/zenodo.20726461, URL: <https://doi.org/10.5281/zenodo.20726461>. law-physical-ai-trials/tree/main/enactment.
- [8] K. Kawchak, "H. r. 9510 (bill v5.0) 2026: Verification before generation in physical ai oncology trials," June 2026. DOI: 10.5281/zenodo.20619762, URL: <https://doi.org/10.5281/zenodo.20619762>. Also single-prompt-bill/tree/main/auto-bill-02.
- [9] K. Kawchak, "H. r. 9510 (bill v4.0) 2026," June 2026. DOI: 10.5281/zenodo.20576907, URL: <https://doi.org/10.5281/zenodo.20576907>. single-prompt-bill/blob/main/auto-bill-01.
- [10] K. Kawchak, "Code generation competition: 16 proprietary vs. open-source llms and iterative learning based on fda adverse event reporting system," Dec. 2025. DOI: 10.5281/zenodo.18029100, URL: <https://doi.org/10.5281/zenodo.18029100>.
- [11] K. Kawchak, "Ai peer review acceleration of llm-generated glioblastoma clinical trial patient matching ml, fda/ich/iso, and fastapi," Dec. 2025. DOI: 10.5281/zenodo.17774560, URL: <https://doi.org/10.5281/zenodo.17774560>.
- [12] K. Kawchak, "Llm-generated glioblastoma drug synergy machine learning: From rapid code prototypes to project deliverables package," Nov. 2025. DOI: 10.5281/zenodo.17614396, URL: <https://doi.org/10.5281/zenodo.17614396>.

- [13] K. Kawchak, “End-to-end oncology clinical trial llm efficiency for industry adoption with fda/ich regulations,” Oct. 2025. DOI: 10.5281/zenodo.17451709, URL: <https://doi.org/10.5281/zenodo.17451709>.
- [14] K. Kawchak, “Accelerating fda compliance and cost efficiency of in silico clinical trials via ai digital twin pancreatic cancer simulation,” Oct. 2025. DOI: 10.5281/zenodo.17239510, URL: <https://doi.org/10.5281/zenodo.17239510>.
- [15] K. Kawchak, “Qsp metastatic pancreatic cancer ai clinical trial simulation from protocol to prediction: Code, vvuq, and playbook,” Aug. 2025. DOI: 10.5281/zenodo.17001137, URL: <https://doi.org/10.5281/zenodo.17001137>.
- [16] K. Kawchak, “Chatgpt 100,000 patient 24-month in silico phase iii 5-arm pancreatic cancer clinical trial triplicate,” July 2025. DOI: 10.5281/zenodo.16415815, URL: <https://doi.org/10.5281/zenodo.16415815>.
- [17] K. Kawchak, “End-to-end pancreatic ductal adenocarcinoma digital twin clinical trial proposals,” June 2025. DOI: 10.5281/zenodo.15735068, URL: <https://doi.org/10.5281/zenodo.15735068>.
- [18] K. Kawchak, “10 year glioblastoma clinical trial meta-analyses by autonomous ai at scale,” May 2025. DOI: 10.5281/zenodo.15549830, URL: <https://doi.org/10.5281/zenodo.15549830>.
- [19] K. Kawchak, “Ai revolution toward the cure of lung adenocarcinoma,” Apr. 2025. DOI: 10.5281/zenodo.15278152, URL: <https://doi.org/10.5281/zenodo.15278152>.
- [20] K. Kawchak, “Autonomous llm agent and scalable reasoning llm for generating cancer drug industry cost solutions,” Mar. 2025. DOI: 10.5281/zenodo.15072843, URL: <https://doi.org/10.5281/zenodo.15072843>.
- [21] K. Kawchak, “Cost containment of global monoclonal antibody drugs and cancer clinical trials via llm focused reasoning,” Feb. 2025. DOI: 10.5281/zenodo.14968404, URL: <https://doi.org/10.5281/zenodo.14968404>.
- [22] K. Kawchak, “Clinical decision support based on bevacizumab cancer trials and pushing the limitations of advanced llms,” Jan. 2025. DOI: 10.5281/zenodo.14968162, URL: <https://doi.org/10.5281/zenodo.14968162>.
- [23] ChatGPT 5.5 Thinking Extended, “Oncology trial long-document decision gates (doctypes).” physical-ai-oncology-trials/trial-documents/research/document-types/ChatGPT-5-5-Thinking-Extended-DocTypes-2026-06-26.md, June 2026. Repository research source. <https://github.com/kevinkawchak/physical-ai-oncology-trials/tree/main/trial-documents/research/document-types>.
- [24] Gemini 3.1 Pro, “Oncology trial long-document steps and phase transitions (doctypes).” physical-ai-oncology-trials/trial-documents/research/document-types/Gemini-3-1-Pro-DocTypes-2026-06-26.md, June 2026. Repository research source. <https://github.com/kevinkawchak/physical-ai-oncology-trials/tree/main/trial-documents/research/document-types>.

- [25] ChatGPT 5.5 Thinking Extended, “Phase 1 oncology document creation workflow (industry workflow).” `physical-ai-oncology-trials/trial-documents/research/industry-workflow/ChatGPT-5-5-Thinking-Extended-Workflow-2026-06-26.md`, June 2026. Repository research source. <https://github.com/kevinkawchak/physical-ai-oncology-trials/tree/main/trial-documents/research/industry-workflow>.
- [26] Gemini 3.1 Pro, “Phase 1 oncology data collection and document creation (industry workflow).” `physical-ai-oncology-trials/trial-documents/research/industry-workflow/Gemini-3-1-Pro-Workflow-2026-06-26.md`, June 2026. Repository research source. <https://github.com/kevinkawchak/physical-ai-oncology-trials/tree/main/trial-documents/research/industry-workflow>.
- [27] International Council for Harmonisation, “Structure and content of clinical study reports e3,” tech. rep., ICH, 1995. <https://www.ich.org/page/efficacy-guidelines>.
- [28] International Council for Harmonisation, “Development safety update report e2f,” tech. rep., ICH, 2010. <https://www.ich.org/page/efficacy-guidelines>.
- [29] International Council for Harmonisation, “Good clinical practice e6(r3),” tech. rep., ICH, 2025. <https://www.ich.org/page/efficacy-guidelines>.
- [30] TransCelerate BioPharma, “The common protocol template: A step toward harmonization,” *Therapeutic Innovation and Regulatory Science*, vol. 51, no. 3, pp. 313–315, 2017. DOI: 10.1177/2168479017696266.
- [31] S. Sharma, “Medical writing in clinical research: A comprehensive guide,” *Perspectives in Clinical Research*, vol. 11, no. 3, pp. 114–118, 2020. DOI: 10.4103/picr.PICR_70_20.
- [32] Electronic Code of Federal Regulations, “21 cfr part 312 - investigational new drug application,” 2026. <https://www.ecfr.gov/current/title-21/chapter-I/subchapter-D/part-312>.
- [33] National Institutes of Health, “Protocol templates for clinical trials,” 2024. <https://grants.nih.gov/policy-and-compliance/policy-topics/clinical-trials/protocol-template>.
- [34] U.S. Department of Health and Human Services, “Hhs launches department-wide effort to restore american leadership in clinical trials,” June 2026. Accessed 2026-06-28. <https://www.hhs.gov/press-room/hhs-launches-clinical-trials-reform-initiative.html>.
- [35] Anthropic, “Introducing claude opus 4.8,” May 2026. Accessed 2026-06-28. <https://www.anthropic.com/news/claude-opus-4-8>.